

Supplementary Computational Material

Predictor-memory frozen-Jacobian methods, manuscript v0.41

Rodrigo Castro Marín

This document is the English counterpart of the computational supplement accompanying manuscript v0.41. It preserves the numerical breakdowns removed from the main article during editorial compression; no essential proof has been moved here. All times below are archived measurements, not fresh measurements from the repository build. The manuscript itself remains a separate document.

1 Detailed external-comparator timings

Tables S1 and S2 retain the results at $\text{TOL} = 10^{-12}$. The main article summarizes all nine configurations per field. Full-precision CSV files, including the other tolerances, are supplied in `data/reference/p5_m6_results.csv` and `p7_m8_results.csv`. The relative quantities are $\Delta t = 100(t_A/t_B - 1)$ and $\Delta W = 100(W_A/W_B - 1)$ for $\kappa = 1$; negative values favor method *A*. Parentheses contain the IQR, in the same units as the median. The time separation rule is $|\tilde{t}_A - \tilde{t}_B| > 2(\text{IQR}_A + \text{IQR}_B)$; it is descriptive, not a formal statistical test.

Table S1: Comparison P_5/M_6 at $\text{TOL} = 10^{-12}$. Times and IQRs are in milliseconds.

Field	δ	Cycles	P_5 time (IQR)	M_6 time (IQR)	Δt [%]	ΔW [%]
H_1	0.10	1/1	0.1258 (0.0064)	0.1252 (0.0075)	+0.48	+6.78
H_1	0.30	2/2	0.2478 (0.0107)	0.2428 (0.0175)	+2.06	+8.89
H_1	0.60	2/2	0.2495 (0.0151)	0.2423 (0.0124)	+2.97	+8.89
H_2	0.10	1/1	0.1752 (0.0104)	0.1747 (0.0124)	+0.29	+5.91
H_2	0.30	1/1	0.1730 (0.0136)	0.1731 (0.0141)	-0.06	+5.91
H_2	0.60	2/2	0.3514 (0.0263)	0.3362 (0.0186)	+4.52	+7.37
H_3	0.10	1/1	0.2442 (0.0389)	0.2506 (0.0310)	-2.55	+3.97
H_3	0.30	1/1	0.2411 (0.0127)	0.2499 (0.0207)	-3.52	+3.97
H_3	0.60	2/2	0.4778 (0.0132)	0.4909 (0.0275)	-2.67	+4.98
H_4	0.10	1/1	0.5495 (0.0301)	0.5704 (0.0699)	-3.66	+3.08
H_4	0.30	1/1	0.5505 (0.0356)	0.5772 (0.0552)	-4.63	+3.08
H_4	0.60	2/2	1.1382 (0.1175)	1.1402 (0.1064)	-0.18	+3.84
H_5	0.10	1/1	1.0378 (0.0843)	1.0821 (0.1272)	-4.09	+2.44
H_5	0.30	1/1	1.0760 (0.0742)	1.1164 (0.0804)	-3.62	+2.44
H_5	0.60	2/2	2.0588 (0.1305)	2.1217 (0.2128)	-2.96	+3.05

Table S2: Comparison $P_7/M8$ at $\text{TOL} = 10^{-12}$. Times and IQRs are in milliseconds.

Field	δ	Cycles	P_7 time (IQR)	$M8$ time (IQR)	Δt [%]	ΔW [%]
H_1	0.10	1/1	0.1537 (0.0084)	0.1202 (0.0050)	+27.87	+15.71
H_1	0.30	1/1	0.1558 (0.0102)	0.1220 (0.0108)	+27.70	+15.71
H_1	0.60	2/2	0.3057 (0.0103)	0.2274 (0.0088)	+34.43	+17.94
H_2	0.10	1/1	0.2047 (0.0096)	0.1774 (0.0155)	+15.39	-5.07
H_2	0.30	1/1	0.2128 (0.0139)	0.1808 (0.0169)	+17.70	-5.07
H_2	0.60	1/1	0.2103 (0.0115)	0.1801 (0.0106)	+16.77	-5.07
H_3	0.10	1/1	0.2850 (0.0176)	0.2844 (0.0148)	+0.21	-23.53
H_3	0.30	1/1	0.2873 (0.0162)	0.2856 (0.0116)	+0.60	-23.53
H_3	0.60	1/1	0.2869 (0.0123)	0.2881 (0.0168)	-0.42	-23.53
H_4	0.10	1/1	0.5839 (0.0392)	0.7361 (0.0495)	-20.68	-30.59
H_4	0.30	1/1	0.6107 (0.0744)	0.7729 (0.0663)	-20.99	-30.59
H_4	0.60	1/1	0.5947 (0.0471)	0.7348 (0.1047)	-19.07	-30.59
H_5	0.10	1/1	1.1568 (0.0984)	1.7986 (0.1976)	-35.68	-34.90
H_5	0.30	1/1	1.0842 (0.0818)	1.6504 (0.1198)	-34.31	-34.90
H_5	0.60	1/1	1.0758 (0.0352)	1.6450 (0.1357)	-34.60	-34.90

2 Full sensitivity of the optimal members

Table S3 retains all six control points. The main article displays two representative endpoints. Stability throughout $[1, 4.5]$ follows from affine normalized-cost differences and the unimodality argument, not from this finite list alone.

Table S3: Stationary family optima and 0.5% near-optimal bands.

Field	κ	N_P^*	m_S^*	$10^6 \eta_P^*$	$10^6 \eta_S^*$	Advantage [%]	P/S bands
H_1	1	3	3	172.461715	165.035043	4.50	$\{3\}/\{3\}$
H_1	1.2	3	3	171.409882	164.058504	4.48	$\{3\}/\{3\}$
H_1	1.298	3	3	170.899153	163.584207	4.47	$\{3\}/\{3\}$
H_1	1.7	3	3	168.835587	161.666981	4.43	$\{3\}/\{3\}$
H_1	3.23511	3	3	161.393776	154.741466	4.30	$\{3\}/\{3\}$
H_1	4.45444	3	3	155.934456	149.649474	4.20	$\{3\}/\{3\}$
H_5	1	9	10	0.555516	0.543469	2.22	$\{8, 9, 10, 11\}/\{9, 10, 11\}$
H_5	1.2	9	10	0.554944	0.542925	2.21	$\{8, 9, 10, 11\}/\{9, 10, 11\}$
H_5	1.298	9	10	0.554664	0.542659	2.21	$\{8, 9, 10, 11\}/\{9, 10, 11\}$
H_5	1.7	9	10	0.553518	0.541570	2.21	$\{8, 9, 10, 11\}/\{9, 10, 11\}$
H_5	3.23511	9	10	0.549186	0.537452	2.18	$\{8, 9, 10, 11\}/\{9, 10, 11, 12\}$
H_5	4.45444	9	10	0.545794	0.534226	2.17	$\{8, 9, 10, 11\}/\{9, 10, 11, 12\}$

3 Secondary graphical representations

Figures S1 and S2 preserve the secondary comparisons. All labels are in English; the underlying measurements are unchanged. A configuration range is not an IQR or confidence interval. No interpolation establishes a universal time crossover or monotonicity.

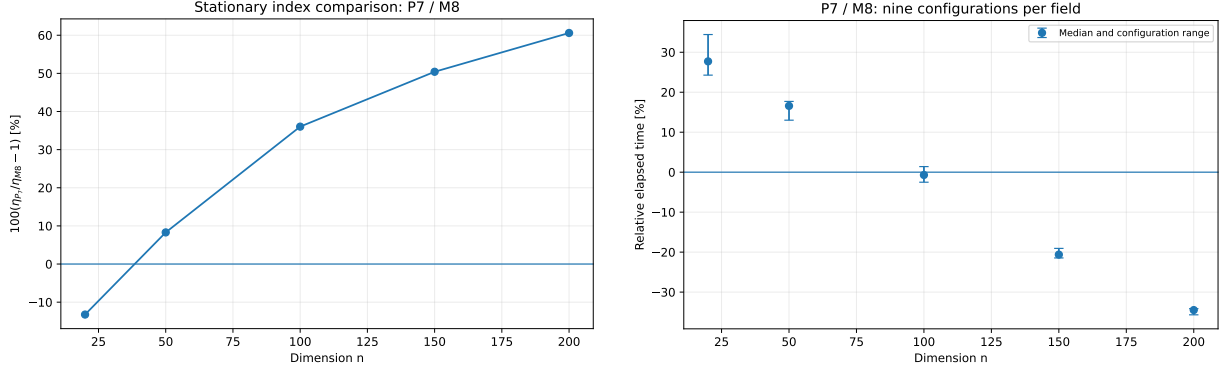


Figure S1: P_7/M_8 : stationary index advantage (left) and median with range of relative elapsed time over nine configurations per field (right). Positive index differences and negative time differences favor P_7 . Time-separation decisions are made separately for each configuration.

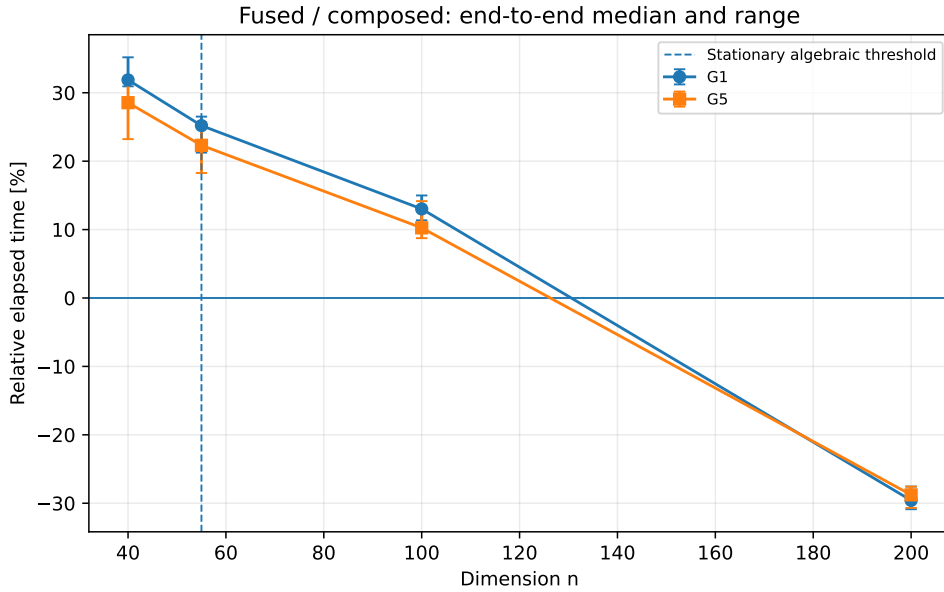


Figure S2: End-to-end comparison of $\hat{P}_2$ with $P_2 \circ P_2$: median and range of relative elapsed time. The line $n = 55$ is the stationary algebraic threshold, not an estimated temporal crossover. Negative values favor fusion.

Provenance note. The archived fusion work columns for G_5 retain a historical evaluation-cost convention described in docs/KNOWN_ISSUES.md. The elapsed-time values in Figure S2 are unaffected. The repository preserves those inputs and provides the separately recomputed canonical costs; no measurement is silently replaced.